

Doctoral Dissertation Review Report**Thesis Title:** Threat Detection in Network Traffic using Time Series Analysis**Author:** Josef Koumar**Institution:** Faculty of Information Technology, Czech Technical University in Prague**Reviewer:** Roberto Magán Carrión, Associate Professor at University of Granada**1. Up-to-Dateness of the Dissertation**

The dissertation addresses a highly critical and rapidly evolving challenge in modern network security: the operational "visibility gap" caused by the near-universal adoption of high-grade encryption standards (such as TLS 1.3, DoH, and ECH). As traditional Deep Packet Inspection (DPI) becomes obsolete due to privacy constraints and computational bottlenecks in core ISP networks environments, the search for non-intrusive traffic analysis methods has become paramount and difficult to address, even more in the context of the detection of suspicious communications behaviors.

The fundamental timeliness of this research lies in its explicit shift away from the prevalent state-of-the-art (SOTA) paradigm: the Sequence of Packet Lengths and Times (SPLT). While traditional SOTA architectures apply heavy Deep Learning methods (e.g., 1D-CNN or LSTM) to SPLT vectors, they are inherently limited to short, localized packet sequences (the first 'n' packets of a single flow) and fail under high-speed operational, computational and storage constraints. By abstracting network telemetry into a structured Time Series Analysis (TSA) framework, the author introduces a cutting-edge multidimensional paradigm. This approach surpasses current SOTA limitations directly through the application of the thesis's core three-tiered taxonomy: single flow, flow and aggregated flow TSA.

Proposed advances in comparison with state-of-the-art main solutions

1. **Single Flow Time Series:** Traditional SOTA requires storing and processing long, raw packet-level SPLT matrices through resource-heavy neural networks. The author's NetTiSA Flow method instead transforms packet streams (flows) into continuous temporal signals and extracts a compact matrix of descriptive statistical-based features (e.g., local variance, means, distributions, etc). It means that a single flow is converted into vector of derived features. This achieves identical threat classification accuracy in encrypted traffic while drastically slashing telemetry data volume and computational overhead, allowing for true *line-rate* execution inside network probes. Additionally, the physical resources needed in terms of memory and processing has been tested in real environments which corroborates the suitability of that solutions ready to be deployed in production environments while keeping similar classification performance than SOTA methods.
2. **Flow Time Series:** Traditional methods analyze network flows in complete isolation, remaining blind to distributed and coordinated attacks. The author's multi-flow framework groups logical connections based on host network dependencies and smartly applies the Lomb-Scargle Periodogram. It allows to extract and create new features like peak frequencies and distances that models certain behaviors. This effectively neutralizes artificial timing noise (*jitter*) injected by attackers, successfully uncovering hidden automated communication channels (such as botnet C&C beaconing and stealthy cryptomining) that isolated mono-flow SOTA approaches entirely miss.
3. **Aggregated Time Series:** Classical forecasting models (e.g., standard Holt-Winters) monitor raw volumetric bulk parameters and generate massive numbers of false positives due to a complete lack of semantic understanding. The author's NODS architecture couples advanced macro-baseline modeling (SARIMA) with a pioneering, zero-shot Large Language Model (LLM) triage engine. Rather than computing numbers, the LLM processes the *semantic context* of the statistical outlier (subnet

attributes, calendar fluctuations), effectively automating root-cause interpretation and dramatically mitigating SOC alert fatigue.

In conclusion, this dissertation is exceptionally up-to-date and practical. It successfully elevates the academic frontier from rigid, resource-expensive short-sequence classifications (SPLT) to scalable, long-term, and context-aware network signal processing (TSA) tailored for immediate enterprise and ISP deployment. It is worth highlighting

2. Formal Structure and Organization of the Dissertation

Structurally, the thesis is well-organized, advancing from the foundational problem statement and literature review, through the core mathematical formalization of the network time-series taxonomy, and into targeted chapters dedicated to solving each level of the taxonomy. Finally, key-insights, conclusions and future work have been very well introduced and described to provide a useful summary allowing the reader for having an useful and complete overview of the whole dissertation work.

3. Completion of the Dissertation Objectives

From my point of view, having into account the proposed objectives the author mentioned in the document, these goals have been successfully and comprehensively fulfilled as it is shown in the following, where each one is addressed.

- **Objective 1: Creation and description of network time series.**

The author formalized a three-tiered taxonomy (Single Flow, Flow Series, and Aggregated Series) that mathematically abstracts raw, unstructured network metadata into structured temporal signals.

- **Objective 2: Research, design, and development of threat detection using TSA.**

The candidate developed three highly sophisticated frameworks aligned with his taxonomy: *NetTiSA Flow* for encrypted traffic classification; a non-uniform *Lomb-Scargle* framework for multi-flow periodicity mining; and the *NODS* architecture powered by SARIMA and zero-shot LLM agents for macroscopic anomaly forecasting and context-aware triage.

- **Objective 3: Evaluation of deployability.**

The dissertation comprehensively evaluates deployability across all infrastructure tiers tailored to each taxonomic level:

- *Single Flow (NetTiSA)*: Evaluated inside production network probes, proving it extracts parameters at line-rate speed with a minimal memory footprint.
- *Flow Series (Periodicity)*: Evaluated at the central collector level, demonstrating a massive reduction in telemetry storage sizes and optimized batch processing times.
- *Aggregated Series (NODS + LLM)*: Evaluated at the SOC backend, proving that spacing out heavy SARIMA retraining windows and triggering the LLM exclusively after statistical anomaly detection minimizes API/computational costs.

- **Objective 4: Creation and publication of open datasets.**

Adhering to open-science benchmarks, the author published 10 open-source repositories and 15 benchmark datasets (e.g., *CESNET-TimeSeries24* and *CESNET-CC25*) on Zenodo and GitHub, garnering over 21,000 community downloads.

4. Assessment of the Methods Used in the Dissertation

The technical execution is evaluated below across the three distinct taxonomies proposed in the research.

4.1 Single Flow Time Series: The NetTiSA Flow Framework

- **Methodological Approach:** This micro-level methodology targets intra-flow packet sequences within an individual IP flow tuple. Rather than utilizing raw Sequence of Packet Lengths and Times (SPLT) arrays which overwhelm sensor buffers, the author extracts low-overhead statistical and autoregressive parameters (such as localized variance, skewness, and auto-correlation) directly from packet lengths and arrival timestamps.
- **Strengths:** By compressing raw packet sequences into structured time-series signals, it achieves a magnitude of data reduction. It matches the binary and multi-class classification fidelity of heavy SOTA Deep Learning architectures (like 1D-CNNs and LSTMs) while maintaining a tiny memory footprint capable of line-rate execution within high-speed ISP hardware probes.
- **Limitations:** Because it aggregates data into statistical parameters to achieve high throughput, it discards fine-grained structural packet sequence dynamics. This could theoretically lead to false negatives when attempting to classify highly polymorphic or adversarial malware variants that purposefully alter their initial statistical footprint.

4.2 Flow Time Series: Multi-Flow Periodicity Mining

- **Methodological Approach:** This mid-level methodology shifts focus from individual packets to the behavioral relationship *between* sequential or concurrent connections. Recognizing that network flows are inherently non-uniform (unevenly spaced due to network traffic and routing), the author adapts the *Lomb-Scargle Periodogram* and pairs it with an optimized Autocorrelation Function (ACF) and peak-distance histogram pipeline.
- **Strengths:** The methodology is uniquely resilient to network noise and communication jitter. By converting multi-flow sequences into frequency-domain signals, it uncovers stealthy, low-frequency automated behaviors—such as botnet Command and Control (C&C) beaconing or persistent cryptomining—even when attackers deliberately inject random delays to deceive traditional single-flow SOTA detectors.
- **Limitations:** The computational cost of calculating Lomb-Scargle periodograms increases with the length of the historical time window and the density of host connections. If a monitored host establishes thousands of distinct parallel connections simultaneously, the central flow collector faces significant processing spikes during batch evaluations.

4.3 Aggregated Time Series: The NODS Architecture & LLM Triage

- **Methodological Approach:** Operating at the macro-level, this framework aggregates volumetric bulk statistics (e.g., bits/sec or flows/sec) across entire subnets into evenly spaced intervals. The Network Outlier Detection System (NODS) utilizes Seasonal Autoregressive Integrated Moving Average (SARIMA) models to forecast baseline network traffic while deploying a dual Mean Outlier Detector (MOD) and Differential Outlier Detector (DOD) to capture anomalies. Crucially, the author appends a Zero-Shot Large Language Model (LLM) agent to semantically interpret the statistical outliers based on environmental context (e.g., subnet types, calendar event registries).
- **Strengths:** The architecture perfectly divides labor between traditional statistical models (which process massive numbers efficiently) and LLMs (which process natural language context). This smart hybrid design solves the industry-wide problem of "alert fatigue" by automatically screening out benign, holiday-driven, or routine maintenance traffic spikes, providing human analysts with clear, actionable text explanations instead of raw numerical alerts.
- **Limitations:** SARIMA baseline training is computationally heavy and represents a clear bottleneck if forced to scale dynamically across millions of endpoints concurrently. Furthermore, the DOD component lacks historical state, which can trigger redundant alerts during rapid traffic normalizations. Lastly, introducing non-deterministic zero-shot LLM reasoning into a critical security

pipeline introduces a minor risk of model hallucination, which could mistakenly down-prioritize a legitimate threat if context inputs are ambiguous.

5. Evaluation of the Results and Contributions of the Dissertation

The empirical results and contributions of this dissertation are highly compelling, demonstrating significant scientific depth and immediate industrial utility. The author does not merely present theoretical models; he validates them using extensive real-world network data and actively shares these resources with the broader scientific community. The specific results and contributions are evaluated below across the three taxonomic levels.

5.1 Single Flow Time Series: NetTiSA Flow Results

- **Empirical Performance:** The *NetTiSA Flow* framework was rigorously benchmarked against 15 foundational cybersecurity datasets covering critical categories like IoT malware, Tor masking, and HTTPS brute-forcing. The results prove that compressing raw packet sequences into structured time-series statistical parameters achieves binary and multi-class classification accuracy on par with heavy, resource-intensive SOTA Deep Learning models (such as 1D-CNN and LSTM networks).
- **Key Scientific Contribution:** The primary contribution here is the mathematical proof that low-overhead side-channel statistical features (e.g., localized variance, auto-regression) can substitute for heavy packet-sequence arrays without sacrificing threat detection fidelity. This achieves an immense reduction in necessary telemetry data storage and network export sizes, making encrypted traffic auditing viable at line-rate speed within core ISP probes.

5.2 Flow Time Series: Multi-Flow Periodicity Mining Results

- **Empirical Performance:** Tested against the author's custom, long-term CESNET-CC25 dataset, the Lomb-Scargle and ACF-based framework demonstrated exceptional resilience against communication noise. It successfully classified malicious applications, pinned down persistent cryptomining behavior, and isolated botnet Command and Control (C&C) "heartbeat" intervals with minimal false discovery rates, even when attackers introduced artificial temporal jitter.
- **Key Scientific Contribution:** The author successfully shifted the detection paradigm from isolated mono-flow analysis to multi-flow logical dependency groups. By introducing the Lomb-Scargle Periodogram to network security, the thesis provides a robust mathematical solution to the long-standing problem of non-uniform sampling in network metadata, creating a highly effective blueprint for exposing automated machine-to-machine threats.

5.3 Aggregated Time Series: NODS and LLM Triage Results

- **Empirical Performance:** The Network Outlier Detection System (NODS) was validated using the massive, real-world CESNET-TimeSeries24 dataset, which tracks 548 institutional subnets over an extended period. Paired with a zero-shot Large Language Model (LLM) framework, expert human annotation verified that the system achieves high-fidelity, transparent contextual reasoning. It successfully differentiated between benign, calendar-driven anomalies (e.g., holiday traffic shifts) and actual security threats.
- **Key Scientific Contribution:** This tier delivers a dual contribution. Mathematically, it provides a highly stable macroeconomic baseline forecasting model using SARIMA coupled with dual-differential outlier detectors. Operationally, it introduces a pioneering hybrid architecture for Explainable AI (XAI) in cybersecurity. By dividing labor between statistics (NODS) and semantics (LLM), it solves the industry-wide bottleneck of SOC alert fatigue, delivering human-readable diagnostic reports and automated alert prioritization.

5.4 Global Contribution: Open Science Leadership

Beyond individual algorithmic successes, the author's overarching contribution to the global research community is really impressive. By adhering strictly to Open Science compliance, the dissertation resulted in the release of 10 distinct open-source repositories and 15 integrated datasets (including *CESNET-TimeSeries24*, *CESNET-CC25*, and *CESNET-MINER22*).

6. Remarks, Objections, and Questions for the Defense

6.1 Global Objection

The primary structural compromise is the lack of a unified orchestration strategy. The author treats the three taxonomic levels as independent silos. In an operational ISP deployment, a macro-anomaly flagged at the Aggregated tier should dynamically trigger deep forensic inspection at the Flow and Single Flow tiers. The dissertation misses the opportunity to formalize this native end-to-end feedback loop.

6.2 Questions for the Candidate

1. **Single Flow:** Can a single flow contain multiple simultaneous attack typologies? If so, which label is chosen during the dataset transformation process? Since NetTiSA discards fine-grained packet order, how vulnerable is it to targeted traffic morphing designed to mimic benign statistical distributions?
2. **Flow Series:** You claim critical behavioral patterns are lost when aggregating flows; could you provide concrete examples of these lost behaviors? Why is it impossible to model long-term periodic behaviors using flows?
 - i. The description of the dataset structure and features extracted from the Lomb-Scargle periodogram remains unclear. What are the key differences in feature distributions among the various data typologies across your solutions?
 - ii. In Table 5.5, you benchmark your solution against external works that utilize completely different training pipelines and ML models. To ensure a fair comparison and rule out external variables, wouldn't a formal Design of Experiments (DoE) and ANOVA—keeping the pipeline identical and altering *only* the Feature Engineering method—be required?
3. **Aggregated Series:** What was the empirical rationale behind selecting aggregation windows from 10 minutes to 1 hour, and what exact impact does this choice have on detection performance? Could you detail how the expert evaluation and labeling process for the network anomalies was conducted?
4. **General Evaluation:** Accuracy is a misleading metric for highly imbalanced datasets. Why was the performance not broken down for each specific attack typology to see if the feature engineering fails for certain threat categories? Are these methods capable of mapping the entire progression of an adversary's campaign across frameworks like MITRE ATT&CK or the Cyber Kill Chain?
5. **Deployability:** Given that deployment constraints vary by tier (probe, collector, SOC backend), what is the primary hardware bottleneck if your entire architecture were deployed in a unified, un-siloed manner within a real ISP?

7. Overall Evaluation and Recommendation

This dissertation is a top-tier academic work that successfully bridges mathematical time-series theory with the pragmatic engineering requirements of high-speed network security. The candidate's deep domain expertise is thoroughly validated by a strong track record of peer-reviewed publications (e.g., CNSM, TMA) and significant community adoption of his open datasets.

Based on its exceptional scientific quality and clear practical impact, I strongly and unreservedly **RECOMMEND** this dissertation for defense.

Friday, 12th of June of 2026

Roberto Magán Carrión,
Reviewer of the Thesis

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